Intelligent Complaint Classification Using Localized Transformer Architectures (E-Commerce): Stakeholder Engagement Plan

1. Selected Stakeholder Groups

- **E-commerce Customers**: Specifically targeting consumers whose communication styles, informal language, or spelling errors are poorly represented in standard datasets.
- **Customer Support Personnel:** The immediate end-users who currently rely on manual categorization and will interact with the system's routing outputs.

2. Stakeholder Priorities

E-commerce customers care most about the immediate, accurate routing of critical issues—such as payments, refunds, and product defects—without being misunderstood due to their use of informal language or spelling errors. They prioritize fairness and customer satisfaction. Customer support personnel care primarily about operational efficiency and reducing their repetitive manual triage workload. They require an accountable system designed as a decision-support tool rather than an opaque replacement, ensuring human oversight is preserved for sensitive, high-impact complaints.

3. Potential Harms and Benefits

- **E-commerce Customers:**
  - Potential Harms: The primary risk is exclusion or misclassification because a consumer's specific language, writing style, or communication channel is poorly represented in the source datasets.
  - Potential Benefits: The system can make complaint submission highly accessible by allowing users to communicate naturally through text rather than forcing them into predefined options or keyword-based processes.
- **Customer Support Personnel:**

- Potential Harms: If the model misunderstands complaints containing multiple issues, it will misroute them, ultimately increasing response times and placing an additional workload on support personnel.
- Potential Benefits: Automating classification will reduce repetitive manual triage and allow these workers to apply their human judgment to uncertain or sensitive complaints.

4. Selected Engagement Methods

- E-commerce Customers: Participatory Design (specifically, co-designing the error taxonomy and edge-case generation).
- Customer Support Personnel: User Committees (specifically, continuous operational integration loops).

5. Method Justification

Most participatory design fails because it asks non-technical users to design technical systems. To fix this, the methodology must restrict input strictly to the stakeholders' domain of expertise: natural language generation and workflow integration.

Participatory Design for E-commerce Customers:

A standard survey or focus group will not capture the data required to improve a localized transformer architecture. Participatory Design fits perfectly here because it extracts actual, unpolished communication patterns. E-commerce customers frequently use informal language, make spelling errors, or combine multiple, distinct issues into a single message. In a highly structured participatory design workshop, customers can be presented with misclassified complaints from the existing dataset—such as those sourced from the Bitext or CFPB data—and asked to rewrite them as they naturally would in a live chat or email application. This stress-tests the model's localized language comprehension. The method is appropriate because it directly targets the project's identified limitation: the underrepresentation of diverse writing styles and communication channels. By having customers co-create adversarial examples—for instance, writing a fraud_unauthorized complaint using regional slang—we expose the rigid boundaries of the taxonomy and directly improve the dataset's coverage before deployment.

User Committees for Customer Support Personnel:

A one-off workshop is wholly insufficient for the end-users of a decision-support tool. User Committees provide the necessary structured, ongoing feedback mechanism. Support staff are the only stakeholders qualified to evaluate whether the mapping of the 10 specific categories—such as correctly distinguishing warranty_repair from product_defect—actually reduces manual triage in practice. This method is highly effective because the system is explicitly designed to support human decision-making rather than replace human judgment. The committee would review the model's performance metrics, specifically precision, recall, macro F1, and per-class F1 scores, and map those technical metrics to actual operational bottlenecks. If the evaluation reveals high precision but low recall on subscription_cancel or billing, the user committee can immediately mandate workflow adjustments. This ensures that human oversight is dynamically re-allocated to the model's weakest or most sensitive categories, transforming theoretical accountability into a practical, operational reality.

6. Influence on Dataset and Model Design

This engagement will force immediate structural changes. The Participatory Design outputs will be injected directly into the training data, artificially boosting the representation of marginalized communication styles and forcing a re-evaluation of the stratified sampling. The User Committee’s feedback will dictate adjustments to class weighting; if support staff find the general_inquiry category acts as a dumping ground, the taxonomy will be fundamentally revised. Consent and licensing conditions for all user-generated data must be fully documented before any redistribution or integration.